Md. Rasul Islam Bapary

AI Engineer · Vision, LLM, Speech & Agentic Systems

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Summary

AI Engineer with 3+ years building production computer vision, llm, speech and agentic systems, specializing inlow-resource(bangla) language AI and on-premise deployment. Shipped 12+ live products spanning OCR and identity verification, bengali ASR-driven workflows, conversational AI and document intelligence system. Experienced in LLM post-training and evaluation. Published in Symmetry (2025).

Work Experience

AI Engineer — Softworks Limited, Dhaka

May 2025 - present

HEALTHCARE & CLINICAL AI

- **Clinova** — Developed a digital healthcare platform for electronic prescriptions and automated doctor–patient communication, simplifying prescription management, patient follow-ups and clinical workflows.
- **Smart Referral System** — Deployed a voice-based referral application connecting Upazila Health Complexes to tertiary hospitals nationwide; STT extracts clinical keywords from speech and auto-fills government referral forms, with all models hosted on-premise for data security. [Slides]
- **PharmaOrder** — Developed a medicine ordering platform between wholesalers and retailers, enabling urgent requests and a streamlined review, modify and approve workflow backed by locally deployed AI models. [Demo]
- **Hotel Booking Agent** — Built an LLM-powered reservation assistant that lets front-desk staff find availability, quote corporate, group and public rates, and hold rooms in the PMS with staff approval. [Demo]

SPEECH & CONVERSATIONAL AI

- **CareMate** — Built an emotion-aware AI companion with local and API-based pipelines (Whisper, Emotion2Vec, Gemma4, Qwen3-TTS) for real-time voice conversations, featuring personalized discussions, routine and medication reminders, and social engagement. [Demo]
- **Voice-to-Data Extraction Platform** — Built a full-stack platform that converts Bengali speech into structured English data using Bangla ASR and a local LLM, with a configurable multi-workflow architecture for healthcare referrals, order management and other use cases.
- **Grievance Reporting System** — Built a voice-driven grievance system for Bengali-speaking migrant workers, integrating STT, LLM-based structured extraction, TTS and PostgreSQL ticketing, with a state-driven conversational flow, automated validation and real-time admin monitoring. [Demo]

VISION & DOCUMENT AI

- **VisionPOS** — Designed a medicine recognition pipeline that indexes deep image embeddings for similarity search against a medicine database, with OCR-based text retrieval to verify each match. [Demo]
- **SmartCard Verification** — Developed a system that verifies existing users and registers new ones by scanning card front and back with Bangla/English OCR, MRZ parsing and 2D barcode reading. [Demo]
- **Financial Document Analysis** — Developed a full-stack platform (English & German) that extracts key metrics from structured and unstructured financial documents and generates analysis reports, forecasts, interactive visualizations and automated risk assessments. [Demo]
- **Vehicle Tracking System** — Implemented licence-plate detection and recognition to record vehicle entries for apartment complexes in Dhaka. [Demo]

ML Engineer — Bangladesh Machine Learning Institute, Dhaka

May 2024 - Apr 2025

- **Automated Garment Quality Control** — Designed a detection pipeline for broken stitches, open seams, dirt marks and foreign objects.
- **Garment Measurement** — Developed an API that measures waist, hem, front rise, back rise and inseam of denim pants from images.
- **PostTraining Agent** — Developed Agentic LLM Training and Evaluation System featuring specialized Trainer and Evaluator Agents to automate model fine-tuning using DPO and GRPO datasets and benchmark trained models across multiple evaluation metrics.

ML Engineer(Intern) — NBSoft, Rangpur

Mar 2023 - Nov 2023

- **Crane & Container Detection** — Trained real-time detection models on live camera footage.
- **Pacemaker Detection** — Built a model to detect pacemakers in human chest X-rays.
- **Bone Segmentation** — Implemented bone segmentation on X-ray images of pneumonia patients.
- **Endoscopy Classification** — Researched and built a Local Attention Grouping model to classify gastritis atrophy and intestinal metaplasia.

Publications & Blog

- Hasan, K.; Tuhin, K.A.; **Bapary, M.R.I.**; Doula, M.S.U.; Alam, M.A.; Ahad, M.A.R.; Uddin, M.Z. MMF-Gait: A Multi-Model Fusion-Enhanced Gait Recognition Framework Integrating Convolutional and Attention Networks. Symmetry 2025, 17, 1155. https://doi.org/10.3390/sym17071155
- State Space Models and Mamba from First Principles. https://medium.com/@ai.rasulbapary/state-space-models-and-mamba-from-first-principles-dc005c518dbd

Skills

- **Languages** Python, C++, C, Java, Bash/Shell
- **AI & ML** Machine Learning, Computer Vision, NLP, Generative AI, Reinforcement Learning
- **LLMs & Speech** Large Language Models, RAG, Prompt Engineering, STT, TTS, OCR
- **Frameworks** PyTorch, scikit-learn, OpenCV, LangChain, Transformers, Ollama, vLLM, unsloth
- **Backend & Web** FastAPI, Flask, Streamlit, React.js, Next.js, Vite, Material-UI, HTML5/CSS3, Tkinter
- **Databases** PostgreSQL, MySQL, SQLite
- **Cloud & Compute** Google Cloud, Microsoft Azure, AWS S3, Lightning AI, Anyscale, Mila Cluster
- **DevOps & Tools** Docker, Git, GitHub Actions (CI/CD), Postman, VS Code, PyCharm, Cursor, Antigravity
- **Embedded & IoT** Raspberry Pi, Arduino, Sensors

Education

B.Sc. in Computer Science & Engineering — Begum Rokeya University, Rangpur 2023
CGPA 3.39 / 4.00 · Ranked 7th in class

Random Projects

- Rohingya campcare referral. [Repo]
- TB case reporting. [Repo]
- HR assistant. [Repo]
- WMH segmentation. [Repo]
- Endoscopy img classification. [Repo]
- Persona generation. [Repo]
- WhisperTraining. [Repo]
- GemmaTune. [Repo]
- DETR. [Repo]
- Signature verification. [Repo]
- AI Finance Accountant Agent. [Repo]
- Ollama chatbot. [Repo]
- Text Summarization. [Repo]
- Sentiment Analysis. [Repo]
- 2D/3D pose and 3D mesh generation. [Repo]
- Bolt Nut Recognition with center measurement.
- Classification of Corn Leaf Disease. [Repo]
- Classification of Autistic Children. [Repo]
- Koala Detection. [Repo]
- X-ray Image Generation. [Repo]

References

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- **Md Shafi Ud Doula**
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